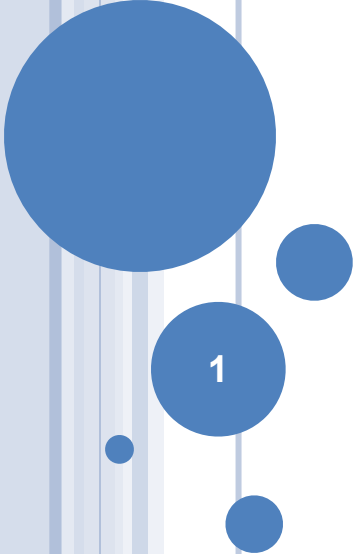


INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 6 The Normal Probability Distribution

1

The left side of the slide features a series of vertical stripes in various shades of blue. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle contains the number '2'.

6.1 PROBABILITY DISTRIBUTIONS FOR CONTINUOUS RANDOM VARIABLES

2

CONTINUOUS RANDOM VARIABLES

- Continuous random variables can assume the infinitely many values corresponding to points on a line interval.
- **Examples:**
 - Heights, weights
 - length of life of a particular product
 - experimental laboratory error

Suppose $X \sim \text{Binomial}(n = 25, p = 0.5)$

Think about

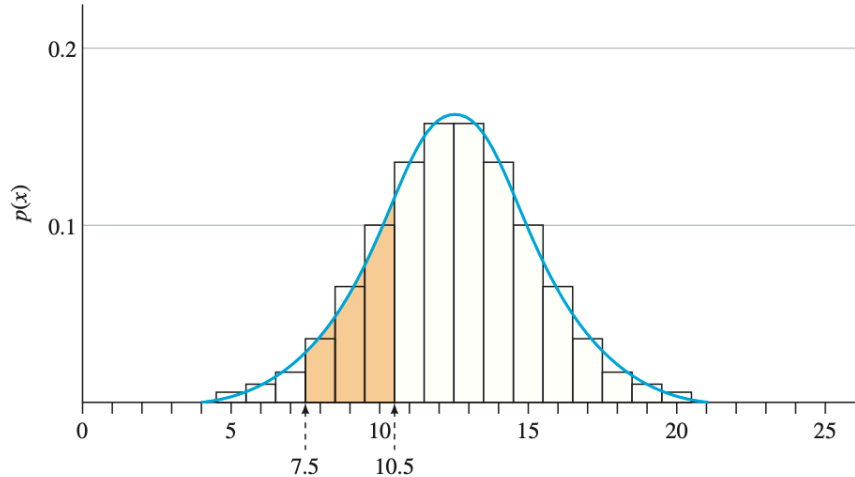
$$P(X \leq 5) = C_0^{25}0.5^{25} + C_1^{25}0.5^{25} + C_2^{25}0.5^{25} + C_3^{25}0.5^{25} + C_4^{25}0.5^{25} + C_5^{25}0.5^{25}$$

using a calculator!

This is very difficult even with a calculator.

Figure 6.17

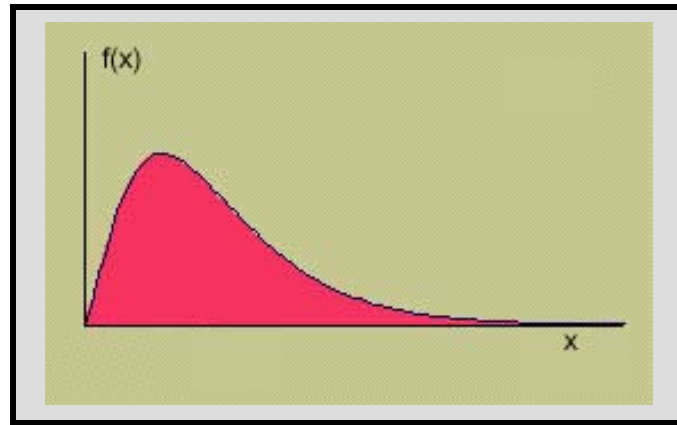
The binomial probability distribution for $n = 25$ and $p = .5$ and the approximating normal distribution with $\mu = 12.5$ and $\sigma = 2.5$



	Discrete RV (Binomial RV)	Continuous RV (Normal RV)
Easy to understand	Yes	No
Easy to calculate its probability	No	Yes

CONTINUOUS RANDOM VARIABLES

- A **smooth curve** describes the probability distribution of a continuous random variable.



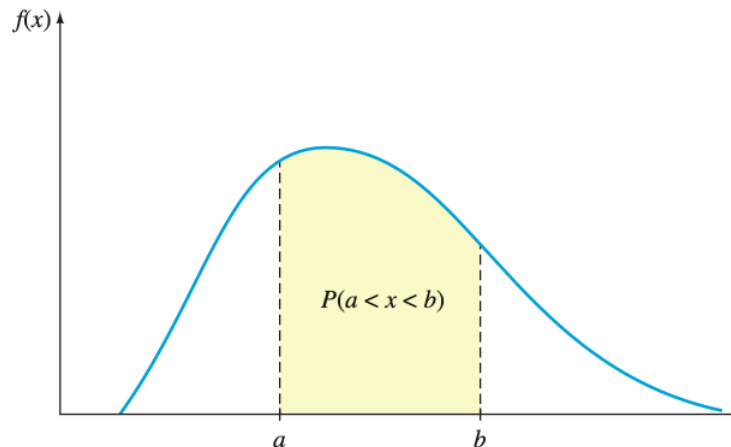
- The depth or density of the probability, which varies with x , may be described by a mathematical formula $f(x)$, called the **probability distribution**, or **probability density function (pdf)** for the random variable x .

PROPERTIES OF CONTINUOUS PROBABILITY DISTRIBUTIONS

- The area under the curve is equal to 1.
- $P(a < x < b) = \text{area under the curve between } a \text{ and } b.$

Figure 6.2

The probability distribution $f(x)$; $P(a < x < b)$ is equal to the shaded area under the curve



- There is no probability attached to any single value of x . That is, **$P(x = a) = 0.$**

MEAN AND VARIANCE OF A CONTINUOUS RANDOM VARIABLE

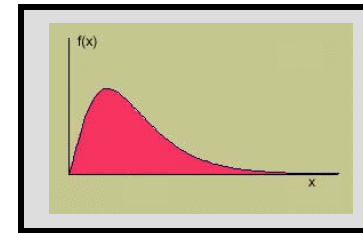
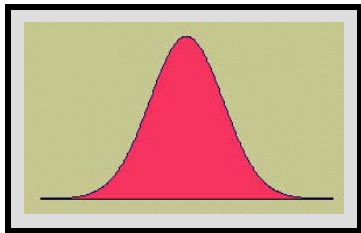
- The mean is $\mu = E[X] = \int xf(x)dx$
- The variance is
$$\sigma^2 = E[(X - \mu)^2] = \int (x - \mu)^2 f(x)dx$$

The left side of the slide features a series of vertical stripes in various shades of blue, ranging from light to dark. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle is particularly large and prominent, while others are smaller and scattered around it.

6.2 THE NORMAL PROBABILITY DISTRIBUTION

8

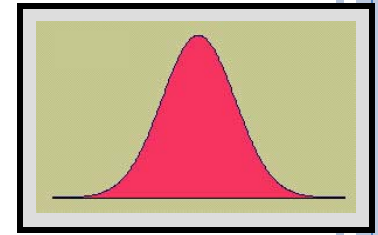
CONTINUOUS PROBABILITY DISTRIBUTION



- There are many different types of continuous random variables
- We try to pick a model that
 - Fits the data well
 - Allows us to make the best possible inferences using the data.
- One important continuous random variable is the **normal random variable**.



THE NORMAL DISTRIBUTION



- The formula that generates the normal probability distribution is:

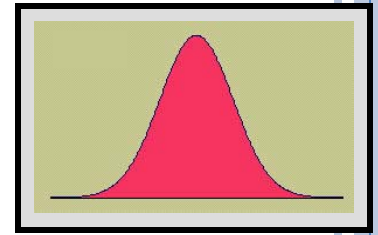
$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \quad \text{for } -\infty < x < \infty$$

$$e = 2.7183 \quad \pi = 3.1416$$

μ and σ are the population mean and standard deviation.

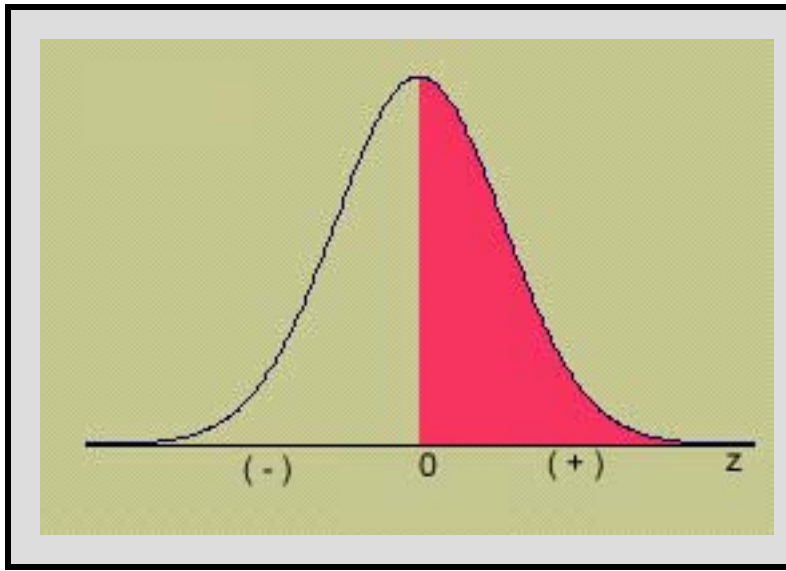
- The shape and location of the normal curve changes as the mean and standard deviation change.

THE STANDARD NORMAL DISTRIBUTION



- To find $P(a < x < b)$, we need to find the area under the appropriate normal curve.
- To simplify the tabulation of these areas, we **standardize** each value of x by expressing it as a z -score, the number of standard deviations σ it lies from the mean μ .

$$z = \frac{x - \mu}{\sigma}$$



THE STANDARD NORMAL (z) DISTRIBUTION

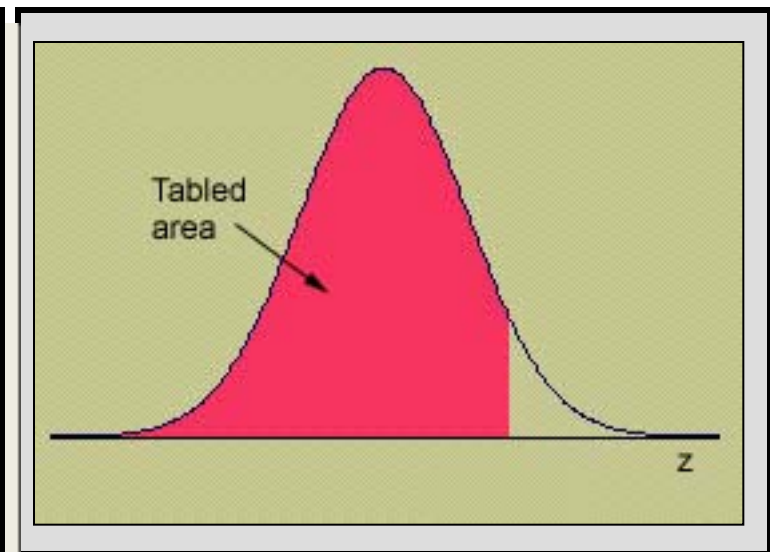
- Mean = 0; Standard deviation = 1
- Symmetric about $z = 0$
- Values of z to the left of center are negative
- Values of z to the right of center are positive
- Total area under the curve is 1.

Table 3 (continued)[illegible]

USING TABLE 3

The four digit probability in a particular row and column of Table 3 gives the area under the z curve to the left that particular value of Z .

z	.00	.01	.02	.03	.04	.05	.06
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278



Area for $z = 1.36$

EXAMPLE

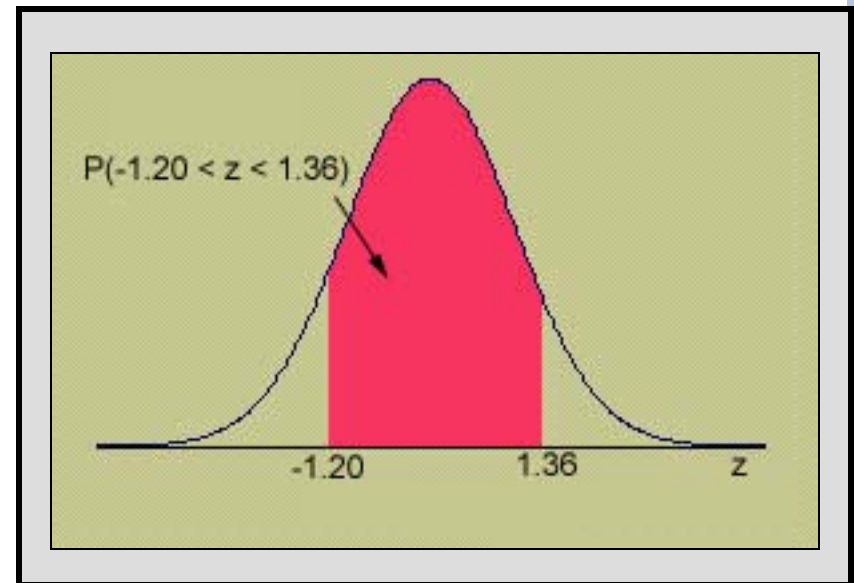


Use Table 3 to calculate these probabilities:

$$P(z \leq 1.36) = .9131$$

$$P(z > 1.36) \\ = 1 - .9131 = .0869$$

$$P(-1.20 \leq z \leq 1.36) = \\ .9131 - .1151 = .7980$$





USING TABLE 3

✓ To find an area to the left of a z-value, find the area directly from the table:

$$P(z < a)$$

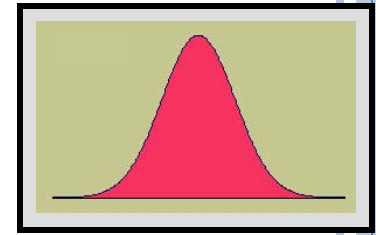
✓ To find an area to the right of a z-value, find the area in Table 3 and subtract from 1:

$$P(z > a) = 1 - P(z < a)$$

✓ To find the area between two values of z, find the two areas in Table 3, and subtract one from the other:

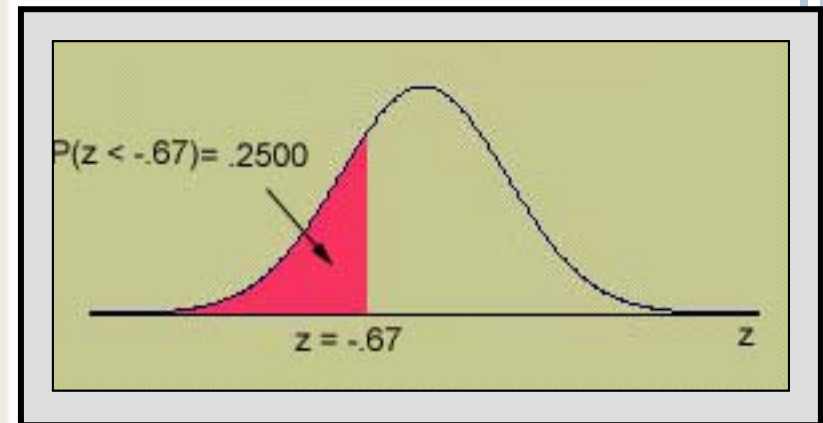
$$P(a < z < b) = P(z < b) - P(z < a)$$

WORKING BACKWARDS



Find the value of z that has area .25 to its left.

1. Look for the four digit area closest to .2500 in Table 3.
2. What row and column does this value correspond to?
3. $z = -.67$

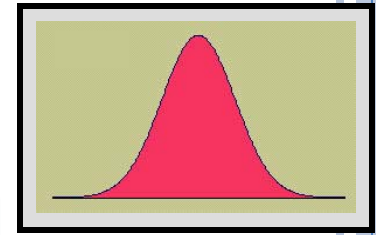


4. What percentile does this value represent?

z	.00	.01	.02	.03	.04	.05	.06	.07	.08
-.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635
-.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894
-.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177
-.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483
-.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810

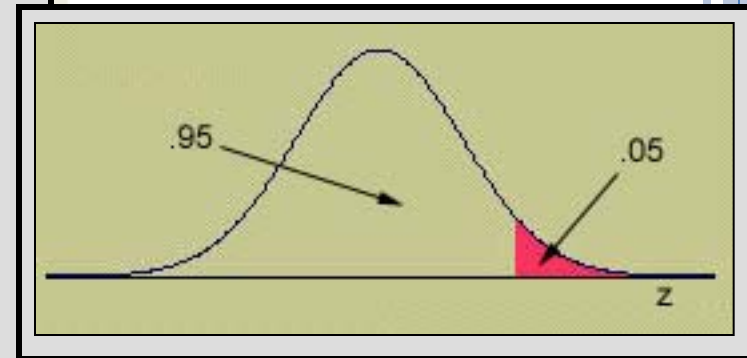
**25th percentile,
or 1st quartile (Q_1)**

WORKING BACKWARDS



Find the value of z that has area .05 to its right.

1. The area to its left will be $1 - .05 = .95$
2. Look for the four digit area closest to .9500 in Table 3.
3. Since the value .9500 is halfway between .9495 and .9505, we choose z halfway between 1.64 and 1.65.
4. $z = 1.645$



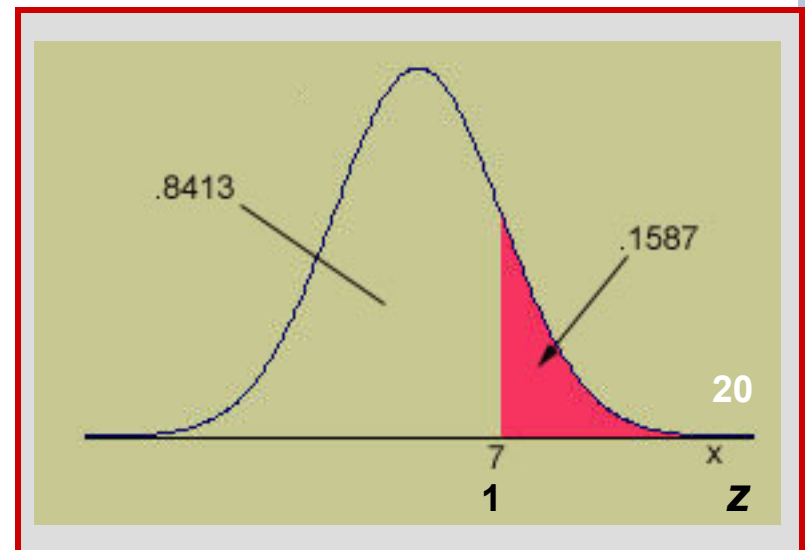
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

FINDING PROBABILITIES FOR THE GENERAL NORMAL RANDOM VARIABLE

- ✓ To find an area for a normal random variable x with mean μ and standard deviation σ , *standardize or rescale* the interval in terms of z .
- ✓ Find the appropriate area using Table 3.

Example: x has a normal distribution with $\mu = 5$ and $\sigma = 2$. Find $P(x > 7)$.

$$\begin{aligned} P(x > 7) &= P\left(z > \frac{7 - 5}{2}\right) \\ &= P(z > 1) = 1 - .8413 = .1587 \end{aligned}$$

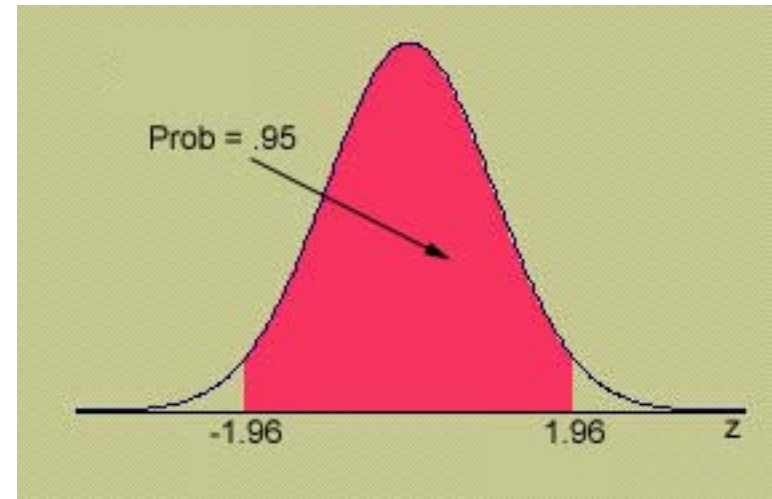




EMPIRICAL RULE

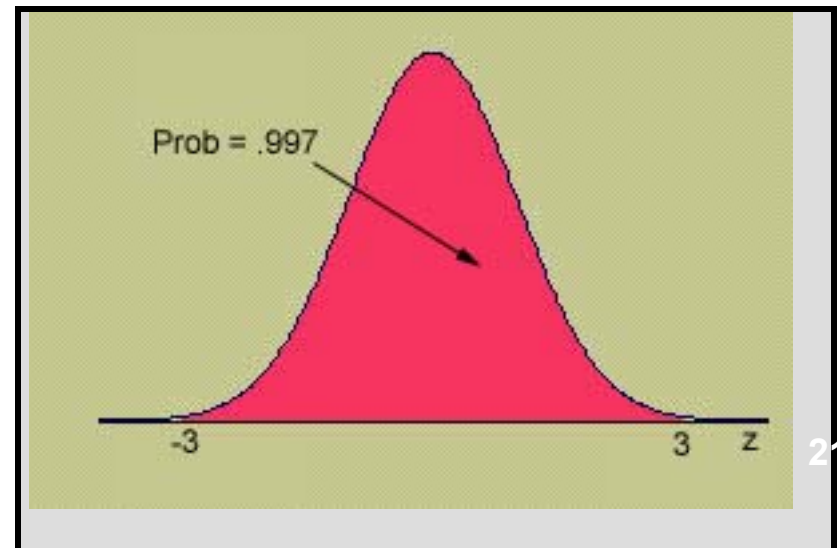
Approximately 95% of the measurements lie within 2 (1.96) standard deviations of the mean.

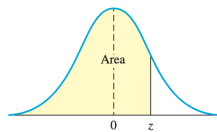
$$\begin{aligned} &P(\mu - 1.96\sigma < x < \mu + 1.96\sigma) \\ &= P\left(\frac{\mu - 1.96\sigma - \mu}{\sigma} < \frac{x - \mu}{\sigma} < \frac{\mu + 1.96\sigma - \mu}{\sigma}\right) \\ &= P(-1.96 < z < 1.96) \\ &= 0.975 - 0.0250 \\ &= 0.9500 \end{aligned}$$



Approximately 99.7% of the measurements lie within 3 standard deviations of the mean.

$$\begin{aligned} &P(\mu - 3\sigma < x < \mu + 3\sigma) \\ &= P\left(\frac{\mu - 3\sigma - \mu}{\sigma} < \frac{x - \mu}{\sigma} < \frac{\mu + 3\sigma - \mu}{\sigma}\right) \\ &= P(-3 < z < 3) \\ &= 0.9987 - 0.0013 \\ &= 0.9974 \end{aligned}$$





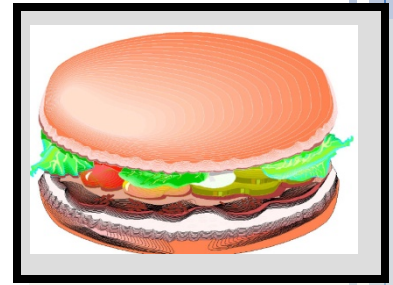
■ Table 3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0007	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

■ Table 3 (continued)

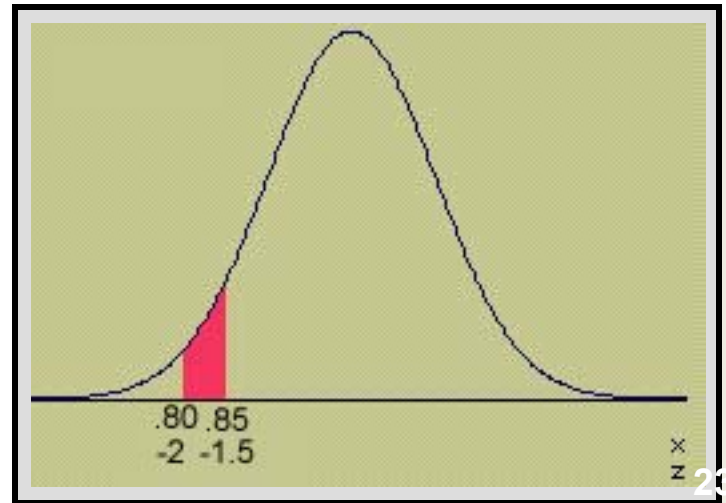
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

EXAMPLE

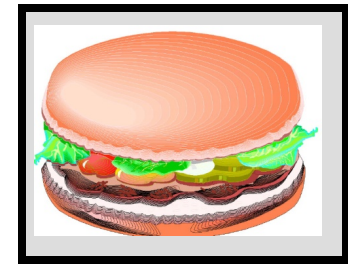


The weights of packages of ground beef are normally distributed with mean 1 pound and standard deviation .10. What is the probability that a randomly selected package weighs between 0.80 and 0.85 pounds?

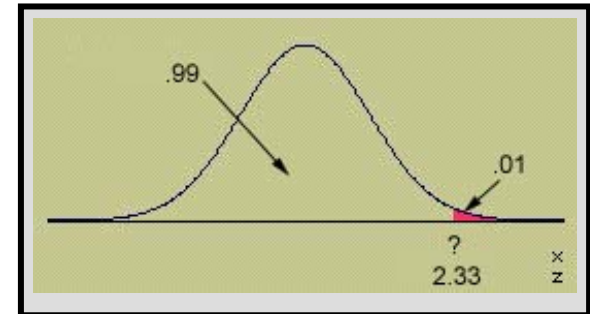
$$\begin{aligned} P(.80 < x < .85) &= \\ P(-2 < z < -1.5) &= \\ .0668 - .0228 &= .0440 \end{aligned}$$



EXAMPLE



What is the weight of a package such that only 1% of all packages exceed this weight?



$$P(x > ?) = .01$$

$$P(z > \frac{?-1}{.1}) = .01$$

From Table 3, $\frac{?-1}{.1} = 2.33$

$$? = 2.33(.1) + 1 = 1.233$$

Table 3 (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9991	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

The slide features a dark blue background with a series of vertical stripes in various shades of blue on the left side. Several blue circles of different sizes are also present, some overlapping the stripes. The main title is written in a white, serif font, centered on the right side of the slide.

6.3 THE NORMAL APPROXIMATION TO THE BINOMIAL PROBABILITY DISTRIBUTION

THE NORMAL APPROXIMATION TO THE BINOMIAL

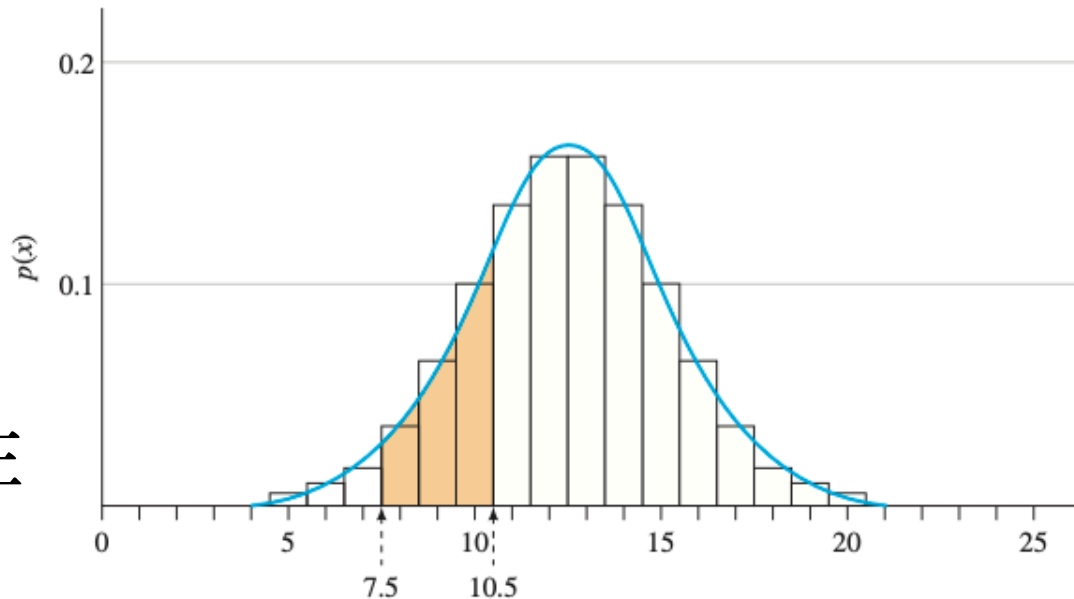
- We can calculate binomial probabilities using
 - The binomial formula: $C_k^n p^k (1 - p)^{n-k}$ for $k = 0, 1, \dots, n$
 - The cumulative binomial tables
- When n is large, and p is not too close to zero or one, areas under the normal curve with mean np and variance npq can be used to approximate binomial probabilities:

$$\text{Binomial}(n, p) \approx N(\mu = np, \sigma^2 = npq)$$

MOTIVATIONS: 連續校正 (CONTINUITY CORRECTION, CC)

Figure 6.17

The binomial probability distribution for $n = 25$ and $p = .5$ and the approximating normal distribution with $\mu = 12.5$ and $\sigma = 2.5$



連續校正

$$P(6 \leq X < 8) \approx P(5.5 < X < 7.5)$$

$$P(6 \leq X \leq 8) \approx P(5.5 < X < 8.5)$$

$$P(X < 5) \approx P(X < 4.5) = P\left(\frac{X - 12.5}{2.5} < \frac{4.5 - 12.5}{2.5}\right) = P(Z < \text{hello})$$

$$P(X \leq 5) \approx P(X < 5.5) = P\left(\frac{X - 12.5}{2.5} < \frac{5.5 - 12.5}{2.5}\right) = P(Z < h2)$$

RULE OF THUMB

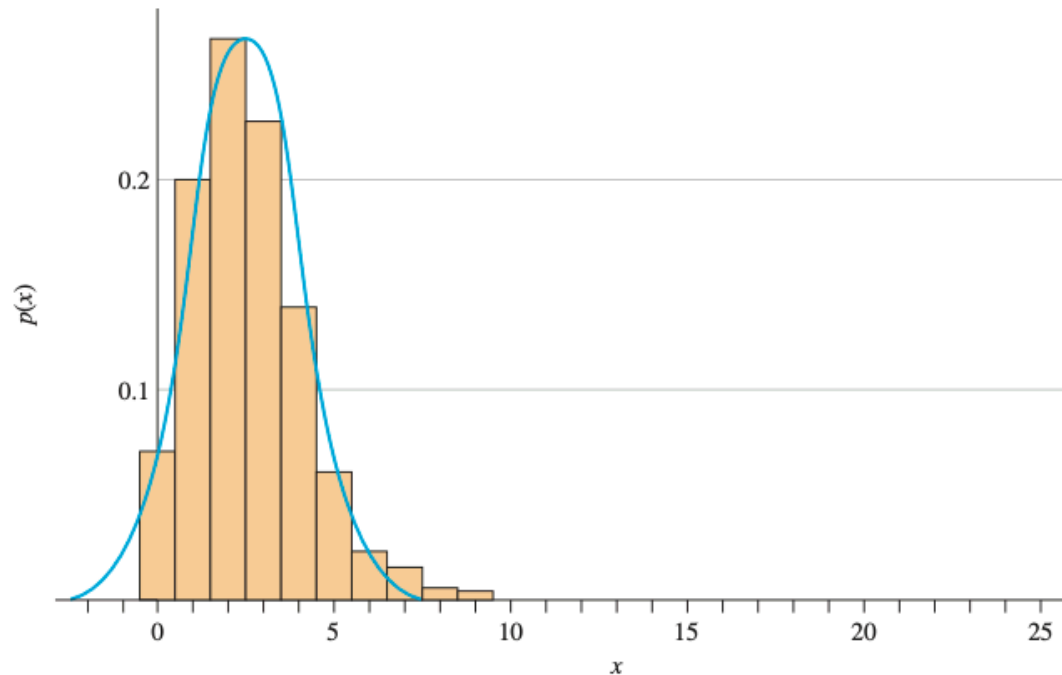
Rule of Thumb

The normal approximation to the binomial probabilities will be adequate if both

$$np > 5 \text{ and } nq > 5$$

Figure 6.18

The binomial probability distribution and the approximating normal distribution for $n = 25$ and $p = .1$



HOW TO CALCULATE BINOMAIL PROBABILITIES USING THE NORMAL APPROXIMATION

1. Find the necessary values of n and p .
2. Make sure $np > 5$ and $nq > 5$
3. Calculate $\mu = np$ and $\sigma^2 = npq$
4. Write the probability you need in x and locate the approximate area on the curve
5. Continuity correct for x by ± 0.5
6. Convert the necessary x -values to z -values using
$$z = \frac{x \pm 0.5 - np}{\sqrt{npq}}$$
7. Use Table 3 in Appendix I to calcualte the approximate probabiltiy.

EXAMPLE

Suppose x is a binomial random variable with $n = 30$ and $p = .4$. Using the normal approximation to find $P(x \leq 10)$.

$$n = 30 \quad p = .4 \quad q = .6$$

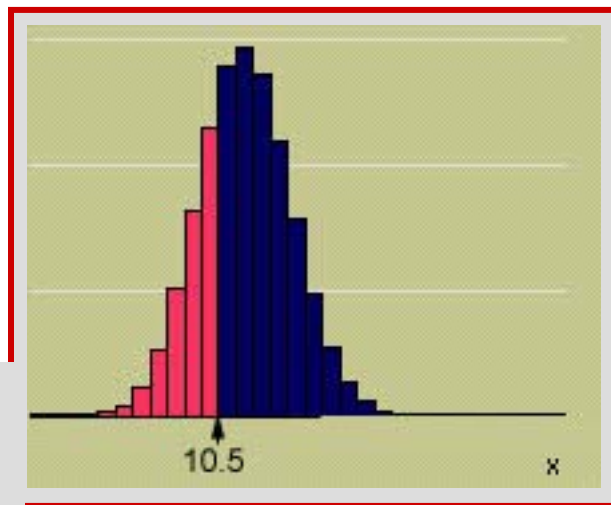
$$np = 12 \quad nq = 18$$

**The normal
approximation
is ok!**

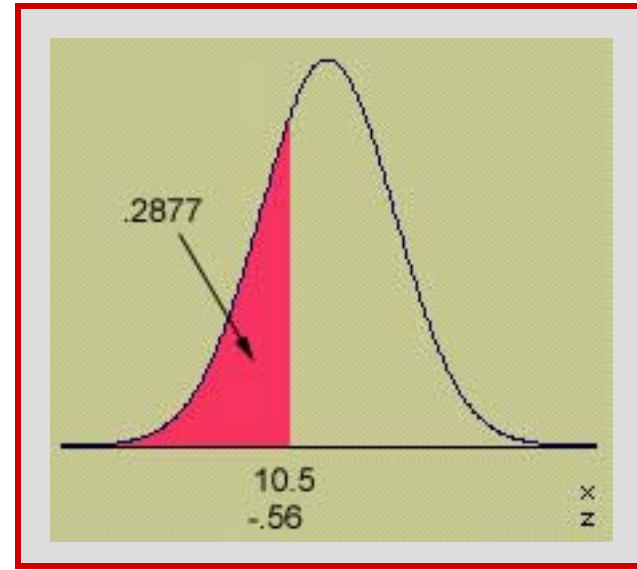
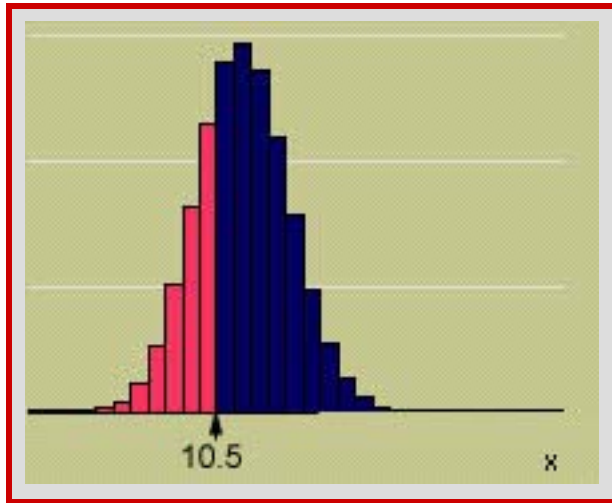
Calculate

$$\mu = np = 30(.4) = 12$$

$$\sigma = \sqrt{npq} = \sqrt{30(.4)(.6)} = 2.683$$



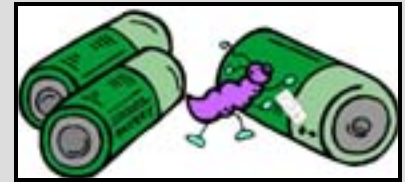
EXAMPLE



$$\begin{aligned} P(X \leq 10) &\approx P(X < 10.5) \\ &= P\left(\frac{X - 12}{2.683} < \frac{10.5 - 12}{2.683}\right) \\ &= P(Z < -0.56) \\ &= 0.2877 \end{aligned}$$

$$\begin{aligned} P(X > 10) &\approx P(X > 10.5) \\ &= P\left(\frac{X - 12}{2.683} > \frac{10.5 - 12}{2.683}\right) \\ &= P(Z > -0.56) \\ &= 1 - 0.2877 \end{aligned}$$

EXAMPLE



A production line produces AA batteries with a reliability rate of 95%. A sample of $n = 200$ batteries is selected. Find the probability that at least 195 of the batteries work.

Success = working battery

Binomial($n=200$, $p=0.95$)

$n = 200$

We need to find $P(X \geq 195)$

$p = .95$ $np = 190$
 $nq = 10$

The normal approximation is
ok!

$$P(X \geq 195) \approx P(X > 194.5)$$

$$= P\left(\frac{X - 190}{\sqrt{200 \times 0.95 \times 0.05}} > \frac{194.5 - 190}{\sqrt{200 \times 0.95 \times 0.05}}\right)$$

$$= P(Z > 1.46)$$

$$= 1 - 0.9278 = 0.0722$$

EXAMPLE (MOVE FROM CH 7)



The soda bottler in the previous example claims that only 5% of the soda cans are underfilled.

A quality control technician randomly samples 200 cans of soda. What is the probability that more than 10% of the cans are underfilled?

$$P(X > 20) \approx P(X > 20.5)$$

$$X(n=200, p=0.05)$$

$$n = 200$$

S: underfilled can

$$p = P(S) = .05$$

$$q = .95$$

$$np = 10 \quad nq = 190$$

$$\begin{aligned} P(\hat{p} > .10) \\ &= P\left(z > \frac{.10 - .05}{\sqrt{\frac{.05(.95)}{200}}}\right) = P(z > 3.24) \\ &= 1 - .9994 = .0006 \end{aligned}$$

This would be very unusual, if indeed $p = .05$!

OK to use the normal approximation

LINK TO HYPOTHESIS TESTING

- Suppose you toss a coin 100 times and got 20 heads. Do you think this is a fair coin? Why or why not?
- 反證法的精神：
- Assume coin is fair $H_0: p = 0.5$
- $P\text{-value} = P(\text{observe this value and more extreme case} \mid p=0.5) = P(X < 20) \sim 0.000$
- Because this probability is so small, we reject the assumption and conclude that the coin is not fair.

KEY CONCEPTS

I. Continuous Probability Distributions

1. Continuous random variables
2. Probability distributions or probability density functions
 - a. Curves are smooth.
 - b. The area under the curve between a and b represents the probability that x falls between a and b .
 - c. $P(x = a) = 0$ for continuous random variables.

II. The Normal Probability Distribution

1. Symmetric about its mean μ .
2. Shape determined by its standard deviation σ .

KEY CONCEPTS

III. The Standard Normal Distribution

1. The normal random variable z has mean 0 and standard deviation 1.
2. Any normal random variable x can be transformed to a standard normal random variable using

$$z = \frac{x - \mu}{\sigma}$$

3. Convert necessary values of x to z .
4. Use Table 3 in Appendix I to compute standard normal probabilities.
5. Several important z -values have tail areas as follows:

Tail Area:	.005	.01	.025	.05	.10
z -Value:	2.58	2.33	1.96	1.645	1.28